

Multiplying and Dividing in Scientific Notation

MA.8.NSO.1.5

I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

Yesterday the powers of ten had to match before you could add.

Today they do not have to match at all — and they can be as far apart as they like.

Warm-Up

On your own. Four minutes. No notes.

1. $(3.4 \times 10^6) + (2.8 \times 10^6)$
2. Write as one power: $\frac{10^9}{10^4}$
3. Find the value: $\sqrt{169}$
4. Find the quotient: $\frac{4.8}{1.6}$

Warm-Up

Answers.

1. $(3.4 \times 10^6) + (2.8 \times 10^6)$ **6.2×10^6**

2. Write as one power: $\frac{10^9}{10^4}$ **10^5**

3. Find the value: $\sqrt{169}$ **13**

4. Find the quotient: $\frac{4.8}{1.6}$ **3**

Notes

Copy both lines.

I. Multiplying: regroup, then use the laws

$$(2 \times 10^4)(3 \times 10^5) = (2 \cdot 3)(10^4 \cdot 10^5) = 6 \times 10^9$$

$$6 \times 10^9, \quad \text{never} \quad 6 \times 10^{20}$$

Multiply the coefficients. Add the exponents — that is the product of powers law, not a new rule.

Notes

II. Dividing: the same move with the quotient law

$$\frac{8.4 \times 10^9}{2.1 \times 10^4} = \frac{8.4}{2.1} \times 10^{9-4} = 4 \times 10^5$$

$$\frac{3 \times 10^8}{3 \times 10^{-4}} = 1 \times 10^{8-(-4)} = 1 \times 10^{12}$$

Divide the coefficients. Subtract the exponents, top minus bottom.

Notes

III. Finish the answer — both directions

$$(8 \times 10^2)(7.5 \times 10^4) = 60 \times 10^6 = 6 \times 10^7$$

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}} = 0.53 \times 10^{-2} = 5.3 \times 10^{-3}$$

Every answer ends with a coefficient at least 1 and less than 10.

Example 1

Multiplying, and the answer needs finishing.

$$(4.5 \times 10^{-3})(6 \times 10^7)$$

Multiply. Write the answer in scientific notation.

Example 1

Worked.

$$(4.5 \times 10^{-3})(6 \times 10^7) = (4.5 \cdot 6)(10^{-3} \cdot 10^7) \quad \text{regroup}$$

$$= 27 \times 10^4 \quad \text{multiply the coefficients, add the exponents}$$

$$27 \times 10^4 = 2.7 \times 10^5 \quad \text{27 is not less than 10: point one place left, exponent up one}$$

Answer: 2.7×10^5

Your Turn

Same steps, your numbers. Boards up when done.

$$(2.5 \times 10^6)(8 \times 10^{-2})$$

Your Turn

Answer.

$$(2.5 \times 10^6)(8 \times 10^{-2})$$

20×10^4 , then repair

Answer: 2×10^5

Example 2

Multiplication and division in one expression.

$$\frac{(8 \times 10^2)(7.5 \times 10^4)}{5 \times 10^2}$$

Evaluate. Write the answer in scientific notation.

Example 2

Worked.

$$(8 \times 10^2)(7.5 \times 10^4) = 60 \times 10^6 \quad \text{the top, regrouped}$$

$$\frac{60 \times 10^6}{5 \times 10^2} = 12 \times 10^4 \quad \text{divide the coefficients, subtract the exponents}$$

$$12 \times 10^4 = 1.2 \times 10^5 \quad \text{finish}$$

Answer: 1.2×10^5

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{(6 \times 10^5)(4 \times 10^{-3})}{8 \times 10^4}$$

Your Turn

Answer.

$$\frac{(6 \times 10^5)(4 \times 10^{-3})}{8 \times 10^4}$$

$$24 \times 10^2 \text{ over } 8 \times 10^4$$

Answer: 3×10^{-2}

Whiteboards • Question 1 of 9

Boards up on three.

$$(3 \times 10^4)(2 \times 10^5)$$

Answer it.

Multiply the fronts, add the backs.

Whiteboards • Question 1 of 9

Answer.

$$(3 \times 10^4)(2 \times 10^5)$$

$$3 \cdot 2 \text{ and } 10^4 \cdot 10^5$$

$$\textbf{Answer: } 6 \times 10^9$$

Whiteboards • Question 2 of 9

Boards up on three.

A printer makes 4×10^3 copies per hour.

It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet.

How many copies does it make?

Answer it.

Rate times time.

Whiteboards • Question 2 of 9

Answer.

A printer makes 4×10^3 copies per hour.

It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet.

How many copies does it make?

10×10^5 , then finish

Answer: 1×10^6

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{9.6 \times 10^8}{1.2 \times 10^3}$$

Answer it.

Top minus bottom.

Whiteboards • Question 3 of 9

Answer.

$$\frac{9.6 \times 10^8}{1.2 \times 10^3}$$

$$9.6 \div 1.2 \text{ and } 10^8 \div 10^3$$

$$\textbf{Answer: } 8 \times 10^5$$

Whiteboards • Question 4 of 9

Boards up on three.

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}}$$

Answer it.

Finish the answer.

Whiteboards • Question 4 of 9

Answer.

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}}$$

0.53×10^{-2} is not finished

Answer: 5.3×10^{-3}

Whiteboards • Question 5 of 9

Boards up on three.

$$(2 \times 10^4)(3 \times 10^5)$$

Which is the product?

A. 6×10^{-1}

B. 6×10^{20}

C. 6×10^9

D. 5×10^9

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$(2 \times 10^4)(3 \times 10^5)$$

Which is the product?

A. 6×10^{-1}

B. 6×10^{20}

C. 6×10^9

D. 5×10^9

Answer: C — 6×10^9

Whiteboards • Question 6 of 9

Boards up on three.

$$(5 \times 10^6)(4 \times 10^3)$$

Which is the product, written in scientific notation?

A. 2×10^9

B. 20×10^9

C. 20×10^{18}

D. 2×10^{10}

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(5 \times 10^6)(4 \times 10^3)$$

Which is the product, written in scientific notation?

A. 2×10^9

B. 20×10^9

C. 20×10^{18}

D. 2×10^{10}

Answer: D — 2×10^{10}

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{3 \times 10^8}{3 \times 10^{-4}}$$

Answer it.

Top minus bottom, and the bottom is negative.

Whiteboards • Question 7 of 9

Answer.

$$\frac{3 \times 10^8}{3 \times 10^{-4}}$$

$$8 - (-4) = 12$$

Answer: 1×10^{12}

Whiteboards • Question 8 of 9

Boards up on three.

$$\frac{8.4 \times 10^7}{n \times 10^2} = 4.2 \times 10^5$$

What is the value of n ?

Answer it.

Look at the coefficients alone.

Whiteboards • Question 8 of 9

Answer.

$$\frac{8.4 \times 10^7}{n \times 10^2} = 4.2 \times 10^5$$

What is the value of n ?

$$8.4 \div n = 4.2$$

Answer: $n = 2$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

For $\frac{7.2 \times 10^9}{2.4 \times 10^4}$ Marcus writes 4.8×10^5

and Sara writes 3×10^{13}

Name what each one did wrong, and give the correct quotient.

Write your answer in sentences.

Each one got exactly one half right.

Whiteboards • Question 9 of 9

Answer.

For $\frac{7.2 \times 10^9}{2.4 \times 10^4}$ Marcus writes 4.8×10^5

and Sara writes 3×10^{13}

Name what each one did wrong, and give the correct quotient.

one wrong in front, one wrong behind

Answer: 3×10^5

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Multiply numbers written in scientific notation — YZU

Divide numbers written in scientific notation — SGT